TASK 3: ADVANCED INTERMITTENT FAULT DIAGNOSTIC INVESTIGATION

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 8 | v1.1

TIME
90 minutes (adapt to workplace conditions)

EVIDENCE
Practical | Knowledge | Documentation

PROGRAMME POINT
Weeks 107-117 | Months 28-31

Your task

Plan and carry out an evidence-led investigation of an intermittent heavy-vehicle electrical, pneumatic or mechanical fault. Capture repeatable evidence and avoid speculative parts replacement.

What you must do

1. Confirm the symptom precisely and gather driver, history and environmental information.
2. Create a staged test plan with expected values and decision points.
3. Reproduce the concern safely and capture scan, measurement or pressure evidence.
4. Analyse correlations, challenge assumptions and isolate root cause.
5. Record the evidence trail, limitations, repair/referral and verification plan.

What good evidence looks like

SAFE CONTROL
Only reproduce faults under approved conditions; stop if testing compromises braking, steering, vehicle control or public safety.

CORRECT METHOD
A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE
Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS
A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle or diagnostic case-study rig
- Technical information and known-good data
- Appropriate scan, electrical or pressure equipment
- Data capture sheet, PPE and job card

SOW ALIGNMENT

Block 8 - Advanced Diagnostic Strategy and Roadside Assistance

KSB evidence focus: K11, S10; reinforced K3, K4, K10, S2-S4, S6, S8-S9, S11, B1-B3

Task reference: HV_L3_Block_8_Task_3_Advanced_Intermittent_Fault_Diagnostic_Investigation

TASK 3: ADVANCED INTERMITTENT FAULT DIAGNOSTIC INVESTIGATION

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Symptom / conditions		Expected value	
Test point / method		Actual result	
Data capture / event		Comparison / correlation	
Root cause confidence		Next test / verification	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why is precise symptom confirmation essential for intermittent faults?

2. What makes a diagnostic test discriminating rather than specula

Knowledge check - continued

3. Explain how you would use known-good data.

4. How do you avoid confirmation bias during diagnosis?

5. What should be recorded if the fault cannot be reproduced?

Assessor/Workplace Mentor observation

- [ ] Builds a logical test plan with decision points.
- [ ] Captures usable, traceable evidence.
- [ ] Records conclusion, confidence and next action.
- [ ] Reproduces the symptom without creating unacceptable risk.
- [ ] Reviews assumptions and avoids unnecessary parts.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- [ ] PASS
- [ ] REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 8 | Task 3

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. ECAS fault

Checks: Check ride-height complaint, air supply/leaks, linkage and sensor condition, DTCs and calibrated height against vehicle data.

Evidence: Record ride-height values, relevant live data, cause and safe verification.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

2. ABS fault

Checks: Verify warning and DTC, then test affected wheel-speed sensor, tone ring, wiring, supply and EBS/ABS data as specified.

Evidence: Record codes, live readings, circuit values and post-repair warning/self-test result.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.